Sagar Gupta

ProServe (Cloud Consultant) - DevOps/MLOps | Amazon Web Services (AWS) | Hyderabad, India

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Summary

MLOps Engineer with 2 years at AWS Professional Services as ProServe (Cloud Consultant) - DevOps/MLOps, building SageMaker ML pipelines and the Terraform and CI/CD that run them. Graduated the AWS Professional Services MLOps SME program by building a SageMaker pipeline for medical image classification, since published by AWS as an official sample: clinical quality and fairness gates, blue/green canary endpoints with auto-rollback, and PSI drift detection that starts retraining, all in Terraform + CodePipeline. Deep IaC and CI/CD foundation from enterprise DevOps engagements. Built LeetCode Rating Predictor and fixed a train/serve skew that had tripled its backtest error (MAE 38.76 to 12.46 on 57 contests). Open-source contributor to Feast, Apache Airflow, and Prefect. 5x AWS certified. LeetCode Guardian – 2166 peak, 1300+ problems solved.

Experience

Amazon Web Services (AWS) Aug 2024 – Present
ProServe (Cloud Consultant) - DevOps/MLOps Hyderabad, India

- **MLOps Pipeline – SME program graduate (Jun 2025 – Aug 2025):** Architected end-to-end SageMaker Pipelines platform for medical image classification – training, model registry, blue-green canary deployment with automatic rollback on error and latency alarms, auto-scaling, and PSI drift detection. Provisioned full infrastructure as code via Terraform + CodePipeline; exposed inference through API Gateway + Lambda with a CloudFront frontend. Generalized and published by AWS as an official open-source sample (aws-samples, MIT-0; see Publications).
- **Lead DevOps Consultant, Enterprise customer (Feb 2026 – Present):** Built a Terraform landing zone: Organizations OUs, Control Tower, SCPs, tag policies, account baselines and least-privilege deploy roles.
  - Automated account vending with Terraform Cloud and OIDC federation.
  - Automated network onboarding and tag-based firewall access with Transit Gateway.
- **DevOps Consultant, Global financial services customer (Oct 2024 – Aug 2025):** Wrote tested Terraform modules (terraform test, PyTest) for a multi-account landing zone. Worked on preventative security controls and EventBridge-based cross-account automation.
- **DevOps Consultant, Financial market infrastructure customer (Aug 2025 – Nov 2025):** Modernized legacy Terraform to cloud-native IaC patterns.
- **Internal Impact:** Published two more AWS samples, on Organizations governance and an AgentCore agent with governed MCP tools (see Publications). AWS technical field community ambassador (AI/ML, DevOps, CloudOps, Agentic AI, Serverless); Anthropic Claude Ambassador (2026). Internal tech talks on Terraform testing and GitOps. Associate Speaker Certification (Renewed 2026); Senior Speaker Certification (2026). Security content review qualification (2026). AWS Jam content tester, DevOps and GenAI (2026).

Amazon Web Services (AWS) Mar 2024 – Jul 2024
Associate Professional Services – DevOps (Internship) Hyderabad, India

- Migrated on-premises application to AWS ECS with blue-green deployment via Terraform, CodePipeline, and CodeDeploy – zero-downtime releases. Instrumented X-Ray and CloudWatch distributed tracing for production observability.

Ikarus-3D May 2023 – Jul 2023
Software Developer Associate (Internship) Mohali, India

- Containerized production application on AWS ECS/ECR with GitHub Actions CI/CD; deployed SonarQube on EC2 + RDS to embed continuous code quality gates into the delivery pipeline.

Publications & Recognition

- **Published by AWS:** aws-samples/sample-sagemaker-image-classification-mlops (author, MIT-0) – SageMaker MLOps in 86 Terraform files across 4 stacks and 13 modules: clinical quality and Fairlearn fairness gates, hourly PSI drift that starts retraining, canary deploys with auto-rollback. *2026*
- **Published by AWS:** aws-samples/sample-aws-terraform-org-governance (author, MIT-0) – Terraform for AWS Organizations governance on Control Tower: six policy types as 20 policy documents, delegated admin for 15 services. *2026*
- **Published by AWS:** aws-samples/sample-agentcore-harness-agent-terraform (author, MIT-0) – AWS architecture agent on the Bedrock AgentCore harness, all in Terraform (71 resources): 3 agent skills, 9 MCP tools behind a default-deny Cedar policy, AgentCore Evaluations gating PROD promotion, CloudWatch alarms and a canary. *2026*
- **LeetCode Guardian** (top 1.1%) – 2166 peak rating, 1300+ problems, 107 contests. GeeksforGeeks 500+ problems; HackerRank 6-Star Problem Solving.

Industry Certifications (5)

AWS: Solutions Architect – Associate | Developer – Associate | Machine Learning Engineer – Associate
AWS: AI Practitioner | Cloud Practitioner

Personal projects and opinions are my own and do not represent Amazon or AWS.

Projects

Ledger Sync Jan 2026
React 19, TypeScript, FastAPI, SQLAlchemy, Neon PostgreSQL, Vercel GitHub | Live
◇ Self-hosted personal-finance app: imports .xlsx, .xls and .csv bank statements with SHA-256 transaction IDs so re-imports stay idempotent, then serves 26 analytics pages (Indian fiscal-year tax planning, investments, FIRE, multi-currency) and an AI assistant limited to 15 read-only tool calls. FastAPI on Vercel over Neon PostgreSQL, React on GitHub Pages, with 298 test files across both stacks.

orrery Oct 2026
Agent Skill, Claude Code Plugin, Node.js, ELK, SVG GitHub | Gallery
◇ Agent Skill that turns a YAML spec into animated architecture diagrams with 18,209 official icons (AWS, Azure, Google Cloud, Kubernetes, GitHub, IBM), each cloud pack rebuilt from the vendor's own download and pinned by checksum. ELK auto-layout, a validator whose every finding names the fix, and a review board of 3 reviewer agents; 40 reference diagrams are rendered and checked in CI. Runs in Claude Code, Codex, Cursor, Gemini CLI and Kiro.

LeetCode Rating Predictor Dec 2023
Python, TensorFlow, NumPy, FastAPI, React, Docker GitHub | Live
◇ Rating model on 15 engineered features. Found and fixed a train/serve skew on participant count that had tripled backtest error (MAE 38.76 to 12.46 on 57 contests). Moved serving to NumPy-only inference pinned by a golden test, cutting the runtime from ~1.8 GB to ~65 MB; a weekly job refreshes the data and re-runs the backtest.

Bedrock Multi-Model MCP Apr 2026
TypeScript, MCP Protocol, AWS Bedrock Converse API GitHub
◇ MCP server putting 26 text models from 7 providers, plus image, video and embedding models, behind 8 tools and one set of AWS credentials. Benchmarked in us-east-1: 26 of 26 models answered and 156 of 156 calls succeeded.

Blue-Green AWS Terraform Jul 2025
Terraform, ECS on EC2, ALB, CodePipeline, CodeDeploy GitHub
◇ 15 Terraform modules for ECS blue/green on an EC2 Auto Scaling Group: 30% canary for 1 minute with rollback on a 5xx spike or unhealthy target, an 80% coverage gate in CodeBuild, and 8 `terraform test` files on a mock provider, plus Checkov and tflint on every PR.

Other Notable Projects: itr-agent (deterministic tax engine, 151 tests), Claude Cost Optimizer (cache-aware Claude cost model), GroundZeroNN (neural network from scratch in NumPy), SelfHub (MCP personal-memory server), Instagram Autopilot (source-grounded animated SVG Reels on a GitHub Actions schedule; Bedrock writes the topic and caption).

Open Source Contributions (11 merged upstream)

- **Apache Airflow** (47K☆) – template_fields on SalesforceBulkOperator, so its arguments render via Jinja at run time (#63109)
- **axios** (109K☆) – AxiosError message now built from AggregateError sub-errors, ending blank dual-stack errors (#11059)
- **lucide** (25K☆) – Fixed a TypeError crash when Vue templates pass the kebab-case icon-node prop (#4512)
- **multer** (12K☆) – Files rejected by fileFilter no longer use up maxCount and fail later valid uploads (#1426)
- **terraform-aws-modules/atlantis** – Fixed custom ECS security group egress rules being silently replaced by allow-all (#446)
- **Also merged:** PrefectHQ/prefect, feast-dev/feast, awslabs/agent-plugins (#132, #212), awslabs/mcp, cloudposse/terraform-aws-tfstate-backend; co-author of the aws-ia/terraform-aws-vpc v4.8.0 fix.

Education

National Institute of Technology Warangal, Telangana Sep 2021 – Jul 2024
Master of Computer Applications (MCA) | NIMCET AIR 208 CGPA: 8.38

Sanskar College of Professional Studies (DAVV University), Indore May 2018 – Aug 2021
Bachelor of Computer Applications (BCA) CGPA: 7.34

Technical Skills

MLOps & ML Platform: SageMaker Pipelines, Model Registry, Endpoints (real-time & async), Custom PSI drift jobs, Blue-Green Deployment, A/B Traffic Shifting, Feature Stores (Feast)
ML Frameworks: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Experiment Tracking, Hyperparameter Tuning
CI/CD for ML: CodePipeline, CodeBuild, GitHub Actions, Docker, Model Versioning, Automated Retraining Triggers
Cloud & Infrastructure: AWS (Lambda, API Gateway, ECS Fargate, CloudFront, EventBridge, S3, CloudWatch, Network Firewall, AWS Backup, Systems Manager Patch Manager, Amazon Bedrock AgentCore), Terraform, HCL, tftest
Data & Serving: FastAPI, GraphQL, REST, Data Pipelines, SHA-256 Deduplication, SQLite, X-Ray, CloudWatch Metrics
Languages: Python, HCL, SQL, TypeScript, JavaScript, Bash

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